10/7/2026

HTML

Full-Stack Development Lab

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Class: BSCS F24 B



Lab No 05

Question 1:

The Interactive Product Configurator should provide a more dynamic and interactive user interface for product customization. Extend the application so that the product controls, customization inputs, messages, and other suitable interface elements are handled using **jQuery**.

The application should support dynamic interaction with product information and customization controls, including real-time updates to the product preview where applicable. Appropriate jQuery-based event handling, element selection, value retrieval, content modification, and DOM manipulation should be demonstrated.

The implementation should provide a clear transition from traditional JavaScript-based interaction to jQuery-based frontend interaction while maintaining the existing functionality of the Product Configurator.

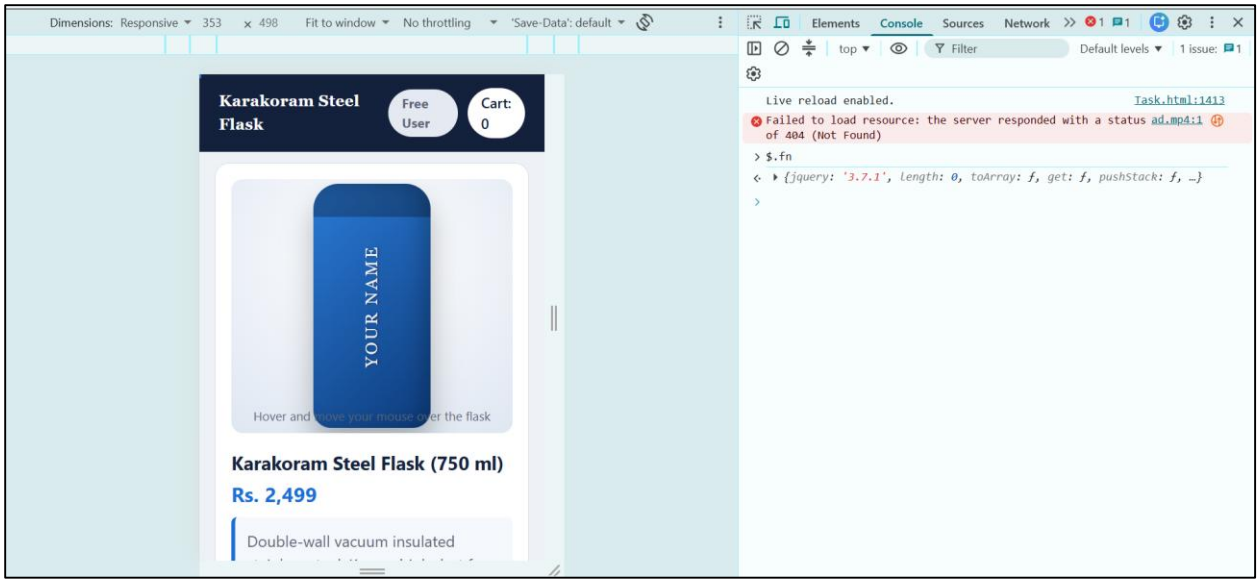
jQuery Implementation:

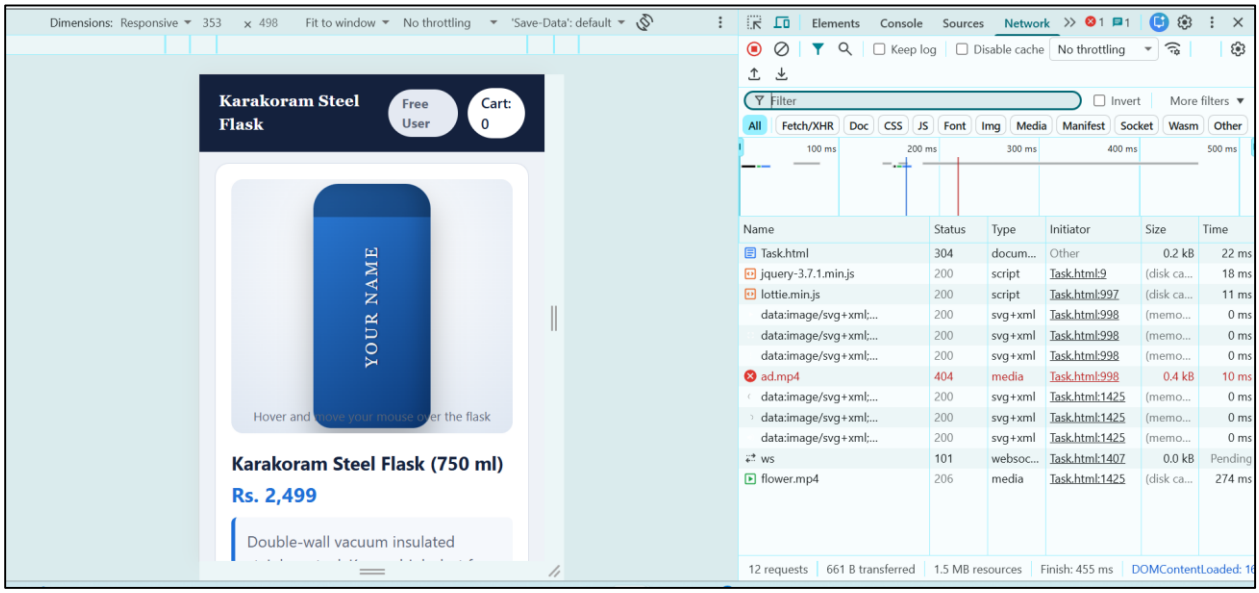
```
<!-- jQuery Library added for Question 1 -->
<script src="https://code.jquery.com/jquery-3.7.1.min.js"></script>
```

```
let isPremium = false;
let activeTheme = null;

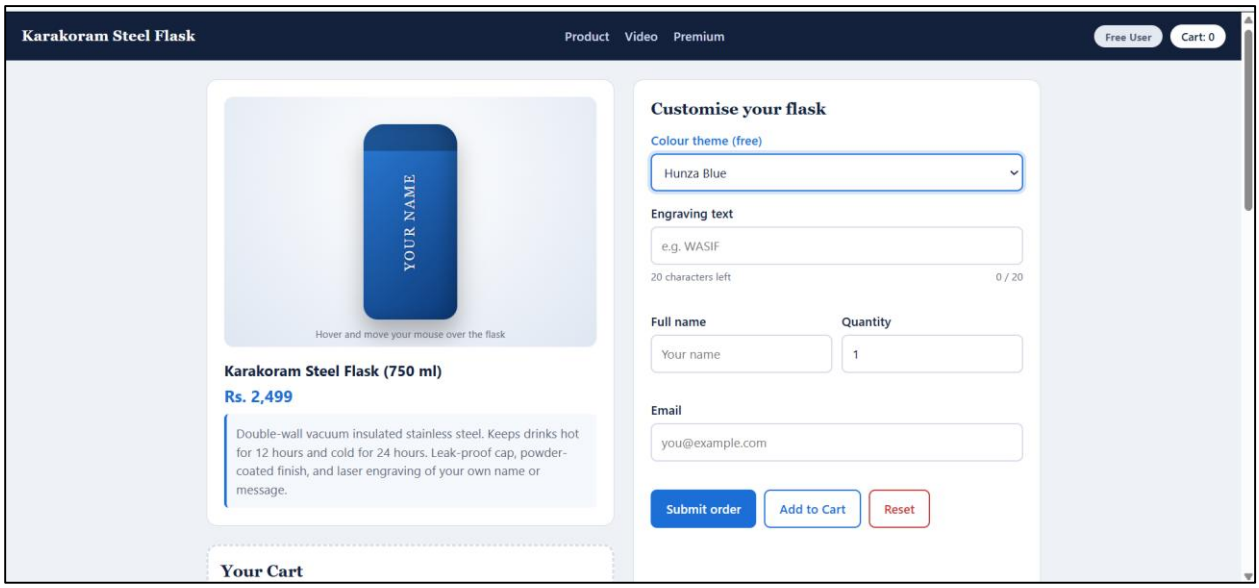
$(document).ready(function () {
    /* ============ HELPERS ============ */
    function showToast(message, type = "") {
        $("#toast").text(message).removeClass().addClass("show " + type);
```

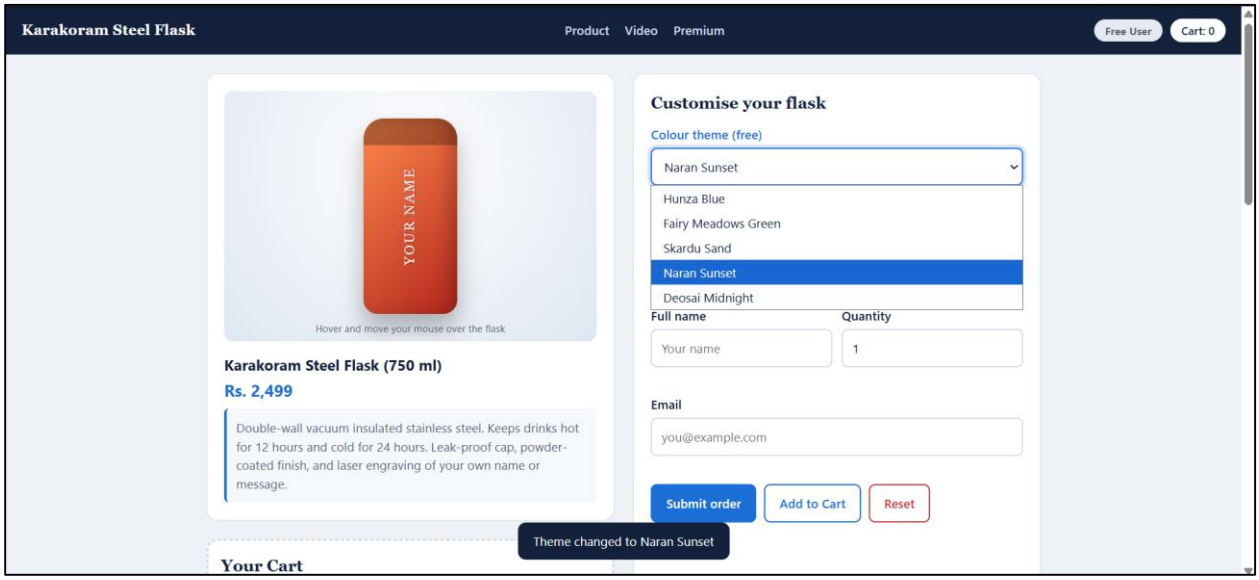
```
/* ============ EVENT: Submit Order ============ */
$("#configForm").on("submit", function (e) {
    e.preventDefault();
    const errors = {};
    const name = $("#fullName").val().trim();
```

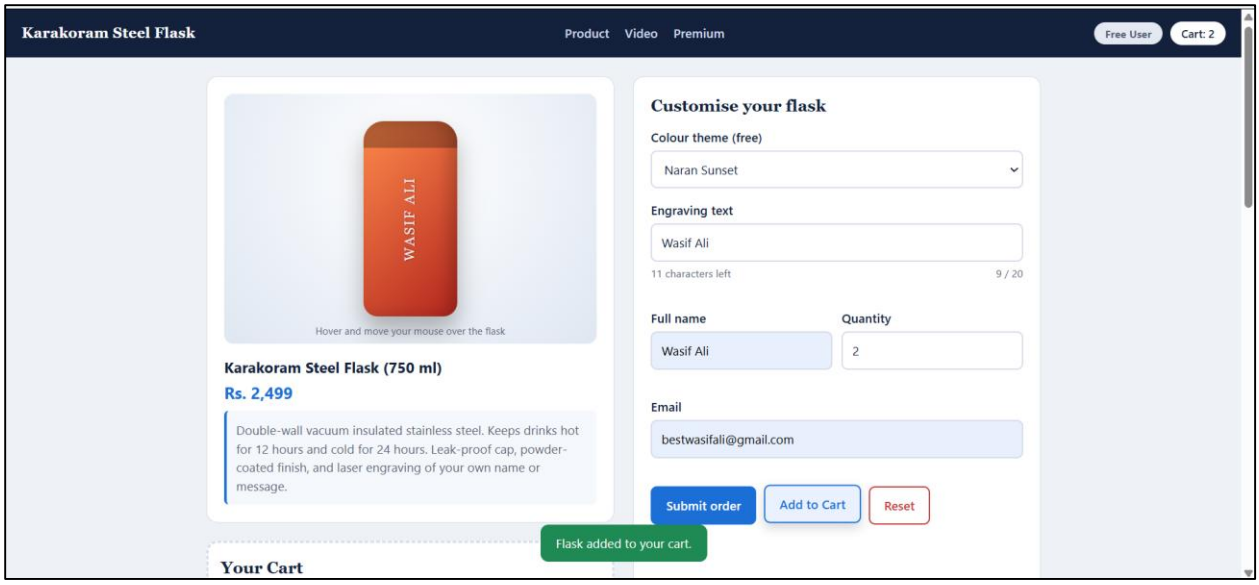

Dimensions: Responsive 353 x 498 Fit to window No throttling 'Save-Data': default
Karakoram Steel Flask
Free User
Cart: 0
YOUR NAME
Hover and move your mouse over the flask
Karakoram Steel Flask (750 ml)
Rs. 2,499
Double-wall vacuum insulated
Elements Console Sources Network
top Filter Default levels 1 issue: 1
Live reload enabled. Task.html:1413
Failed to load resource: the server responded with a status of 404 (Not Found) ad.mp4:1
$.fn
{jquery: '3.7.1', length: 0, toArray: ƒ, get: ƒ, pushStack: ƒ, …}

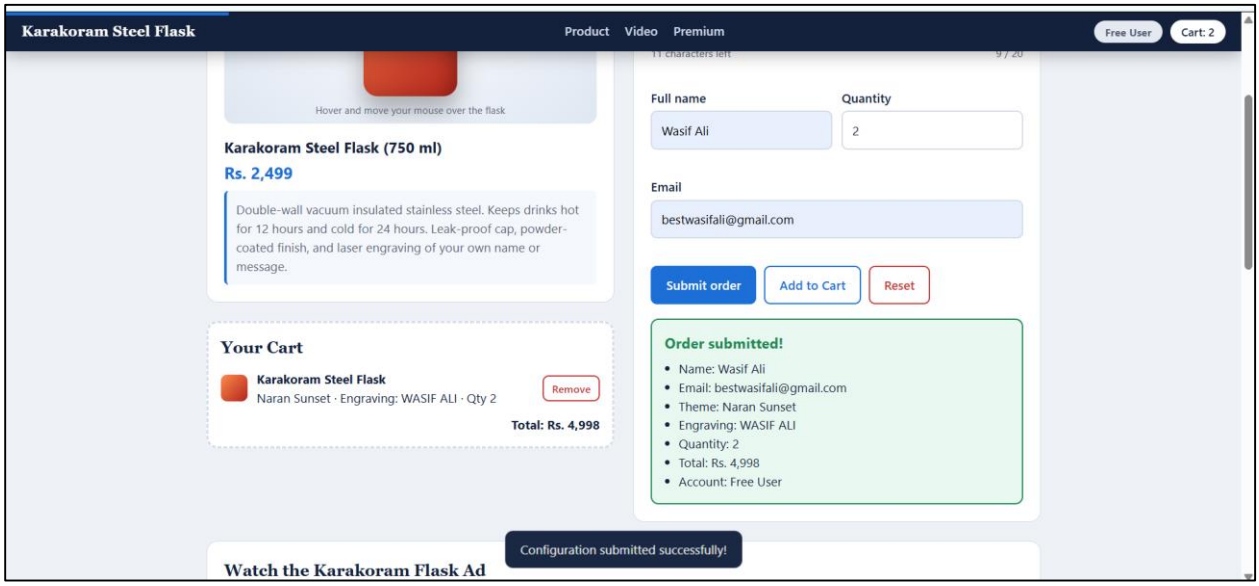

Dimensions: Responsive 353 x 498 Fit to window No throttling 'Save-Data': default
Karakoram Steel Flask
Free User
Cart: 0
YOUR NAME
Hover and move your mouse over the flask
Karakoram Steel Flask (750 ml)
Rs. 2,499
Double-wall vacuum insulated
Elements Console Sources Network
Keep log Disable cache No throttling
Filter Invert More filters
All Fetch/XHR Doc CSS JS Font Img Media Manifest Socket Wasm Other
100 ms 200 ms 300 ms 400 ms 500 ms
Name Status Type Initiator Size Time
Task.html 304 docum... Other 0.2 kB 22 ms
jquery-3.7.1.min.js 200 script Task.html:9 (disk ca... 18 ms
lottie.min.js 200 script Task.html:997 (disk ca... 11 ms
data:image/svg+xml;... 200 svg+xml Task.html:998 (memo... 0 ms
data:image/svg+xml;... 200 svg+xml Task.html:998 (memo... 0 ms
data:image/svg+xml;... 200 svg+xml Task.html:998 (memo... 0 ms
ad.mp4 404 media Task.html:998 0.4 kB 10 ms
data:image/svg+xml;... 200 svg+xml Task.html:1425 (memo... 0 ms
data:image/svg+xml;... 200 svg+xml Task.html:1425 (memo... 0 ms
data:image/svg+xml;... 200 svg+xml Task.html:1425 (memo... 0 ms
ws 101 websoc... Task.html:1407 0.0 kB Pending
flower.mp4 206 media Task.html:1425 (disk ca... 274 ms
12 requests 661 B transferred 1.5 MB resources Finish: 455 ms DOMContentLoaded:

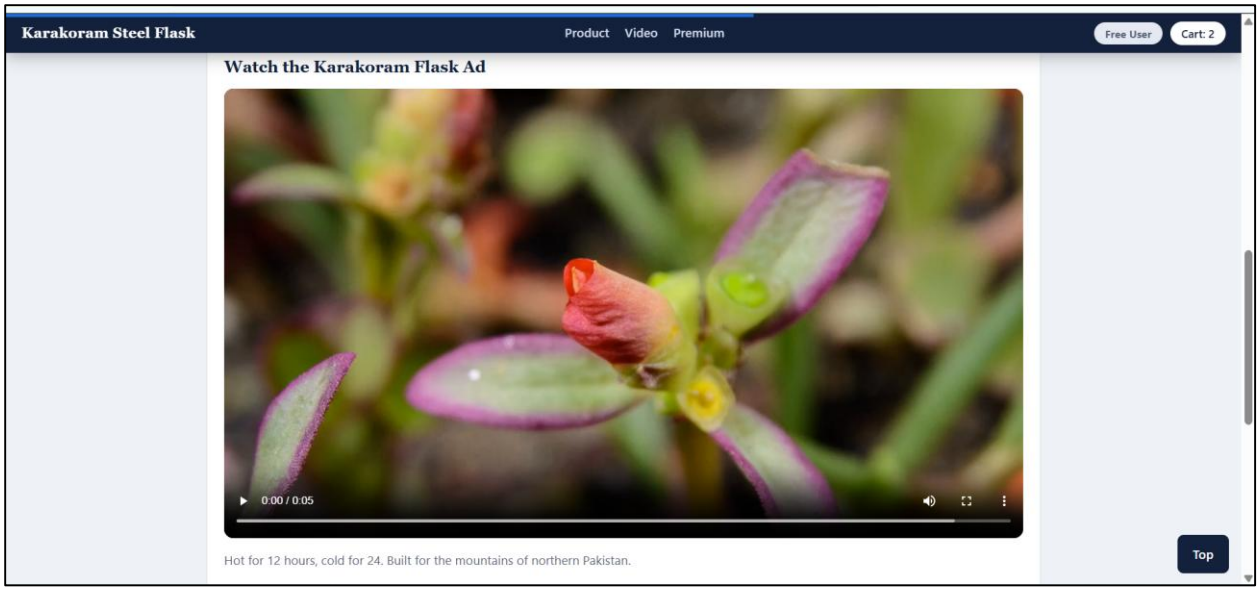
Result:

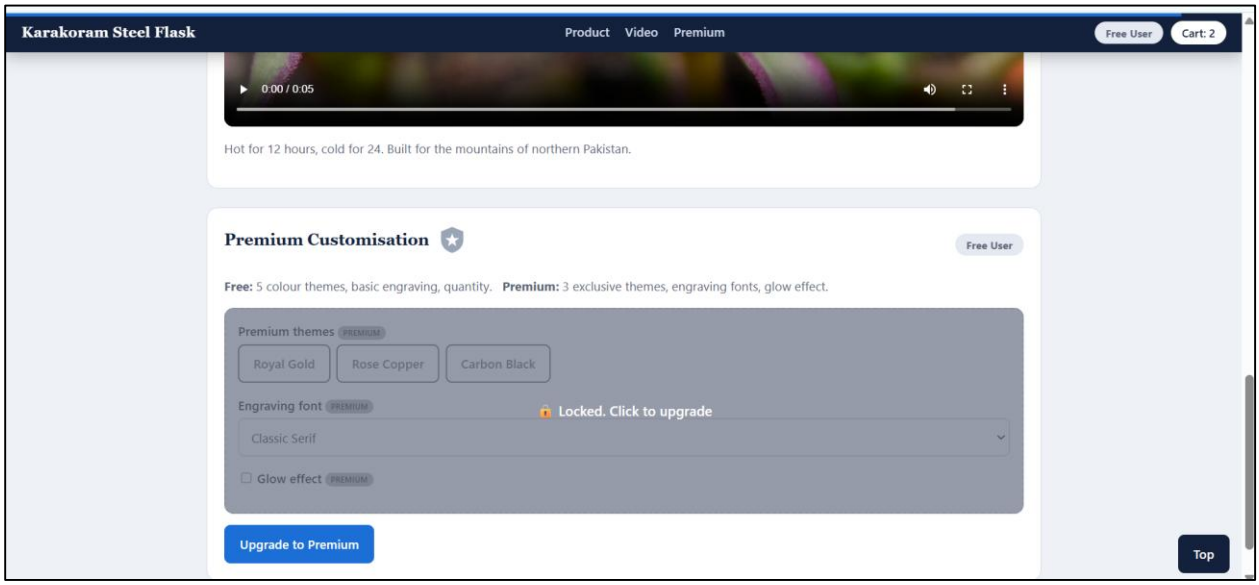

Karakoram Steel Flask
Product
Video
Premium
Free User
Cart: 0
YOUR NAME
Hover and move your mouse over the flask
Karakoram Steel Flask (750 ml)
Rs. 2,499
Double-wall vacuum insulated stainless steel. Keeps drinks hot for 12 hours and cold for 24 hours. Leak-proof cap, powder-coated finish, and laser engraving of your own name or message.
Customise your flask
Colour theme (free)
Hunza Blue
Engraving text
e.g. WASIF
20 characters left
0 / 20
Full name
Your name
Quantity
1
Email
you@example.com
Submit order
Add to Cart
Reset
Your Cart

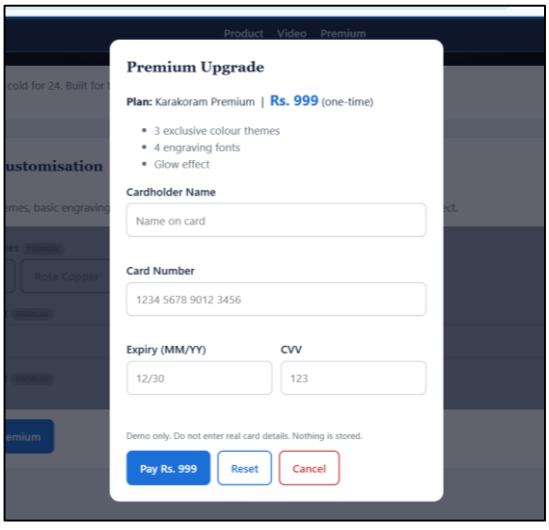

Karakoram Steel Flask
Product
Video
Premium
Free User
Cart: 0
YOUR NAME
Hover and move your mouse over the flask
Karakoram Steel Flask (750 ml)
Rs. 2,499
Double-wall vacuum insulated stainless steel. Keeps drinks hot for 12 hours and cold for 24 hours. Leak-proof cap, powder-coated finish, and laser engraving of your own name or message.
Customise your flask
Colour theme (free)
Naran Sunset
Hunza Blue
Fairy Meadows Green
Skardu Sand
Naran Sunset
Deosai Midnight
Full name
Your name
Quantity
1
Email
you@example.com
Submit order
Add to Cart
Reset
Theme changed to Naran Sunset
Your Cart

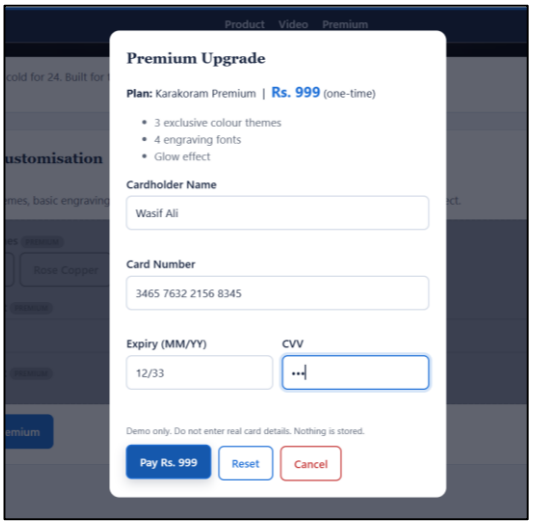

Karakoram Steel Flask
Product
Video
Premium
Free User
Cart: 2
WASIF ALI
Hover and move your mouse over the flask
Karakoram Steel Flask (750 ml)
Rs. 2,499
Double-wall vacuum insulated stainless steel. Keeps drinks hot for 12 hours and cold for 24 hours. Leak-proof cap, powder-coated finish, and laser engraving of your own name or message.
Customise your flask
Colour theme (free)
Naran Sunset
Engraving text
Wasif Ali
11 characters left
9 / 20
Full name
Quantity
Wasif Ali
2
Email
bestwasifali@gmail.com
Submit order
Add to Cart
Reset
Flask added to your cart.
Your Cart

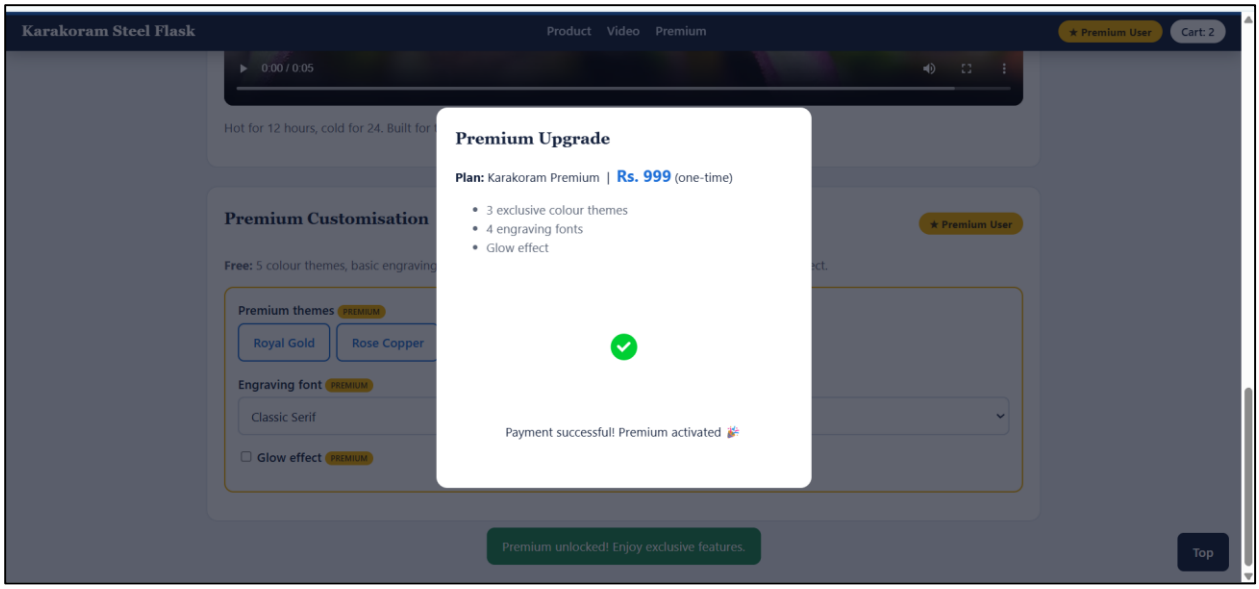

Karakoram Steel Flask
Product
Video
Premium
Free User
Cart: 2
Hover and move your mouse over the flask
Karakoram Steel Flask (750 ml)
Rs. 2,499
Double-wall vacuum insulated stainless steel. Keeps drinks hot for 12 hours and cold for 24 hours. Leak-proof cap, powder-coated finish, and laser engraving of your own name or message.
Full name
Quantity
Wasif Ali
2
Email
bestwasifali@gmail.com
Submit order
Add to Cart
Reset
Your Cart
Karakoram Steel Flask
Naran Sunset · Engraving: WASIF ALI · Qty 2
Remove
Total: Rs. 4,998
Order submitted!
Name: Wasif Ali
Email: bestwasifali@gmail.com
Theme: Naran Sunset
Engraving: WASIF ALI
Quantity: 2
Total: Rs. 4,998
Account: Free User
Configuration submitted successfully!
Watch the Karakoram Flask Ad


Karakoram Steel Flask
Product
Video
Premium
Free User
Cart: 2
Watch the Karakoram Flask Ad
0:00 / 0:05
Hot for 12 hours, cold for 24. Built for the mountains of northern Pakistan.
Top


Karakoram Steel Flask
Product
Video
Premium
Free User
Cart: 2
0:00 / 0:05
Hot for 12 hours, cold for 24. Built for the mountains of northern Pakistan.
Premium Customisation
Free User
Free: 5 colour themes, basic engraving, quantity. Premium: 3 exclusive themes, engraving fonts, glow effect.
Premium themes
Royal Gold
Rose Copper
Carbon Black
Engraving font
Locked. Click to upgrade
Classic Serif
Glow effect
Upgrade to Premium
Top


Premium Upgrade
Plan: Karakoram Premium | Rs. 999 (one-time)
3 exclusive colour themes
4 engraving fonts
Glow effect
Cardholder Name
Name on card
Card Number
1234 5678 9012 3456
Expiry (MM/YY)
12/30
CVV
123
Demo only. Do not enter real card details. Nothing is stored.
Pay Rs. 999
Reset
Cancel


Premium Upgrade
Plan: Karakoram Premium | Rs. 999 (one-time)
3 exclusive colour themes
4 engraving fonts
Glow effect
Cardholder Name
Wasif Ali
Card Number
3465 7632 2156 8345
Expiry (MM/YY)
12/33
CVV
Demo only. Do not enter real card details. Nothing is stored.
Pay Rs. 999
Reset
Cancel



Premium Customisation

★ Premium User

Free: 5 colour themes, basic engraving, quantity. **Premium:** 3 exclusive themes, engraving fonts, glow effect.

Premium themes PREMIUM

Royal Gold | Rose Copper | Carbon Black

Engraving font PREMIUM

Typewriter

☑ Glow effect PREMIUM

Question 2

The Product Configurator should obtain product information from an external source instead of relying entirely on hard-coded product data.

Integrate a suitable **public product API** into the application and retrieve product information asynchronously. The retrieved information may include product name, price, category, image, description, or other relevant product attributes.

The information should be requested when the user performs an appropriate interaction and should be displayed dynamically within the application.

The application should demonstrate the complete communication process between the webpage and the external API, including the request, JSON response, response handling, and dynamic update of the webpage. Appropriate handling should also be provided for unsuccessful requests.

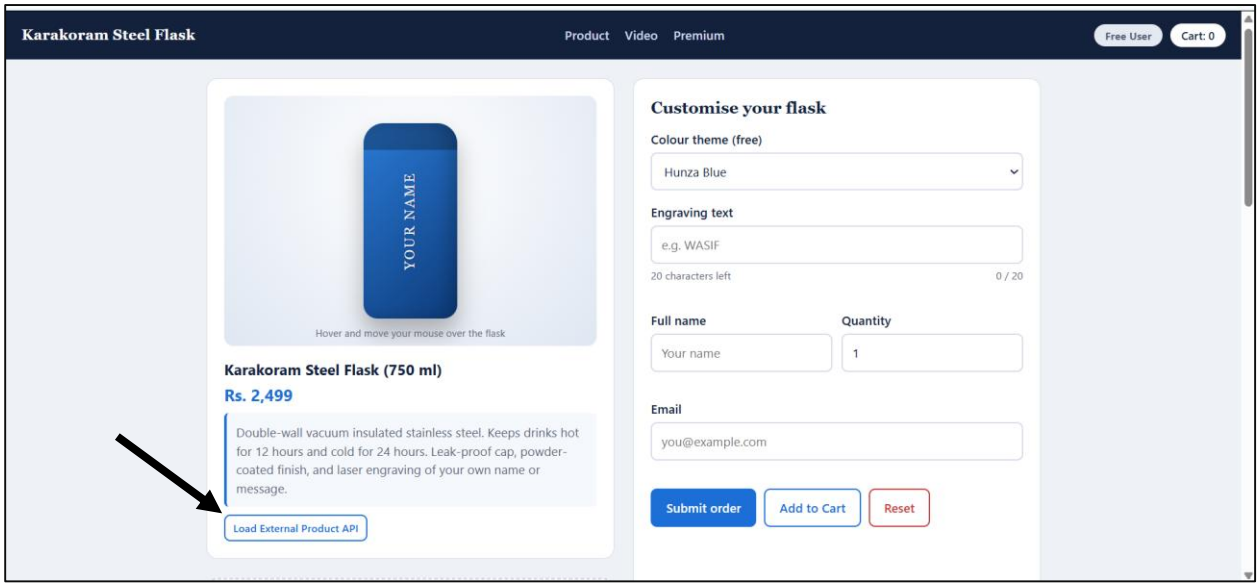
The implementation should use **jQuery AJAX** and demonstrate a **GET request**.

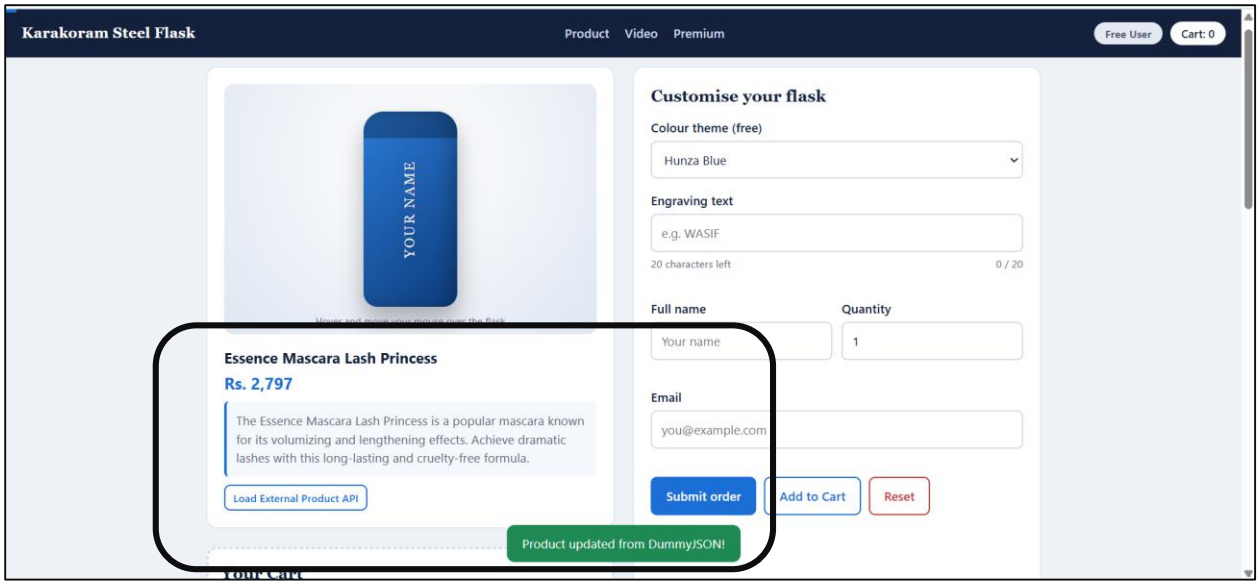
The request and response should be observable through the browser's Developer Tools.

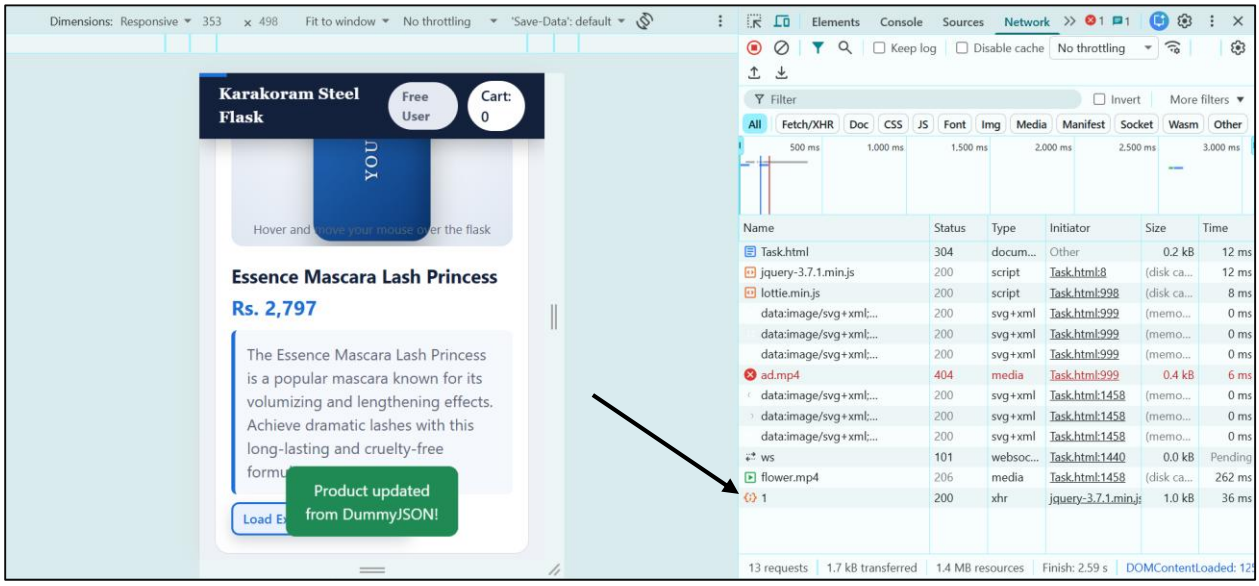
jQuery AJAK Implementation:

```
/* ============ QUESTION 2: EXTERNAL API GET REQUEST ============ */
$("#loadApiBtn").on("click", function () {
    const $btn = $(this);
    $btn.text("Fetching Data...").prop("disabled", true);

    $.ajax({
        url: "https://fakestoreapi.com/products/1",
        method: "GET",
        dataType: "json",
```




Karakoram Steel Flask
Product
Video
Premium
Free User
Cart: 0
YOUR NAME
Customise your flask
Colour theme (free)
Hunza Blue
Engraving text
e.g. WASIF
20 characters left
0 / 20
Full name
Your name
Quantity
1
Email
you@example.com
Essence Mascara Lash Princess
Rs. 2,797
The Essence Mascara Lash Princess is a popular mascara known for its volumizing and lengthening effects. Achieve dramatic lashes with this long-lasting and cruelty-free formula.
Load External Product API
Submit order
Add to Cart
Reset
Product updated from DummyJSON!


Dimensions: Responsive 353 x 498 Fit to window No throttling 'Save-Data': default
Karakoram Steel Flask
Free User
Cart: 0
Hover and move your mouse over the flask
Essence Mascara Lash Princess
Rs. 2,797
The Essence Mascara Lash Princess is a popular mascara known for its volumizing and lengthening effects. Achieve dramatic lashes with this long-lasting and cruelty-free
Product updated from DummyJSON!
Elements
Console
Sources
Network
Keep log
Disable cache
No throttling
Filter
Invert
More filters
All
Fetch/XHR
Doc
CSS
JS
Font
Img
Media
Manifest
Socket
Wasm
Other
Name
Status
Type
Initiator
Size
Time
Task.html 304 docum... Other 0.2 kB 12 ms
jquery-3.7.1.min.js 200 script Task.html:8 (disk ca... 12 ms
lottie.min.js 200 script Task.html:998 (disk ca... 8 ms
data:image/svg+xml;... 200 svg+xml Task.html:999 (memo... 0 ms
data:image/svg+xml;... 200 svg+xml Task.html:999 (memo... 0 ms
data:image/svg+xml;... 200 svg+xml Task.html:999 (memo... 0 ms
ad.mp4 404 media Task.html:999 0.4 kB 6 ms
data:image/svg+xml;... 200 svg+xml Task.html:1458 (memo... 0 ms
data:image/svg+xml;... 200 svg+xml Task.html:1458 (memo... 0 ms
data:image/svg+xml;... 200 svg+xml Task.html:1458 (memo... 0 ms
ws 101 websoc... Task.html:1440 0.0 kB Pending
flower.mp4 206 media Task.html:1458 (disk ca... 262 ms
1 200 xhr jquery-3.7.1.min.j... 1.0 kB 36 ms
13 requests
1.7 kB transferred
1.4 MB resources
Finish: 2.59 s
DOMContentLoaded: 12

Question 3

The Product Configurator should communicate with a locally developed backend for handling application data.

Develop a simple backend using **Node.js and Express.js** and integrate it with the Product Configurator.

The backend should provide suitable product-related information through a **GET API endpoint** using predefined JSON data.

In addition, the application should allow the user to submit a selected product configuration to the backend through a **POST request**. The submitted information may include product name, selected color, variant, size, engraving, or other customization details.

The frontend and backend should exchange the information using JSON. The POST request should demonstrate conversion of the JavaScript object into JSON data, while the backend should receive the submitted information through the request body and return an appropriate JSON response to the frontend.

The backend should also demonstrate the use of **Express middleware**, including JSON parsing and at least one custom middleware capable of observing request information such as the HTTP method and requested URL.

Where the frontend and backend are running on different ports, the application should be configured so that the browser can successfully communicate with the backend.

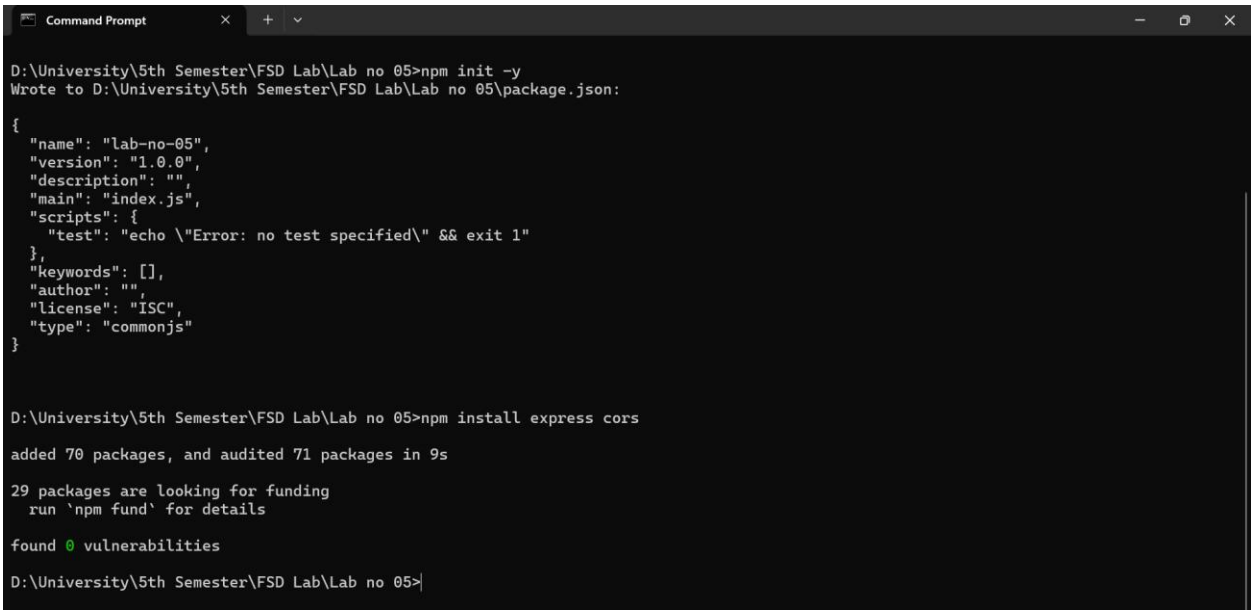
The final implementation should demonstrate both communication flows:

```
Frontend → AJAX GET → Express → JSON Response → Frontend
```

and

```
Frontend → AJAX POST → Express → Request Body → JSON Response → Frontend
```

The request and response information should be observable using the browser Developer Tools and the backend terminal.



```
PS D:\University\5th Semester\FSD Lab\Lab no 05> node server.js
Backend server running at http://localhost:3000
```

Server.js code:

```
const express = require('express');
const cors = require('cors');

const app = express();
const PORT = 3000;

// 1. CORS Middleware (allows your HTML file to talk to this server)
app.use(cors());

// 2. Built-in Middleware for JSON parsing
app.use(express.json());

// 3. Custom Middleware (logs HTTP method and requested URL)
app.use((req, res, next) => {
    console.log(`[LOG] ${req.method} request made to: ${req.url}`);
    next();
});

// GET API Endpoint (Predefined JSON data)
app.get('/api/product', (req, res) => {
    res.json({
        id: 101,
        title: "Karakoram Special Edition",
        description: "Fetched from local Express backend. Premium insulated
flask.",
        price: 2999
    });
});

// POST API Endpoint (Receives the frontend configuration)
app.post('/api/order', (req, res) => {
    const orderData = req.body;

    // Log the received JSON data in the backend terminal"
    console.log("=== New Order Received ===");
    console.log(orderData);

    // Send a JSON response back to the frontend
```

```
    res.json({
        status: "success",
        message: "Order successfully processed by Node.js Backend!",
        received_data: orderData
    });
});

app.listen(PORT, () => {
    console.log(`Backend server running at http://localhost:${PORT}`);
});
```